



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Project Task 1

Programme: Bachelor of Computer Science (Data Engineering)

Subject: SECR1213 Network Communications

Session-Sem: 2024/2025-1

Section: 02

Group name: Router ranger

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Brief Explanation for the Task

The Faculty of Computing (FC) is planning to construct a new two-storey building equipped with modern technology, such as networking devices that support education in line with 4IR standards, to accommodate an anticipated 15% growth in student and academic staff numbers over the next four years. Our group aims to design a cutting-edge and cost-effective network infrastructure that supports growth, ensuring students and staff have access to technology in a comfortable environment for study.

The building should include 2 general-purpose labs, 1 Cisco Network lab, 1 Embedded lab, and a student lounge, each 14m x 10m in size and a total 30 workstations for each lab with equipped a high-speed internet connection. We should design the floor plan of the building with any suitable software based on the requirements on the case study. The network design should feature a high-speed core backbone, secure segmentation, and protocols to protect against network breaches. Additionally, the infrastructure will be scalable and support wireless connectivity, supporting future growth and equipped with technological devices, in line with Faculty Computing's objective to create an innovative and comfortable learning environment.

Suggested Floor Plan

First Floor

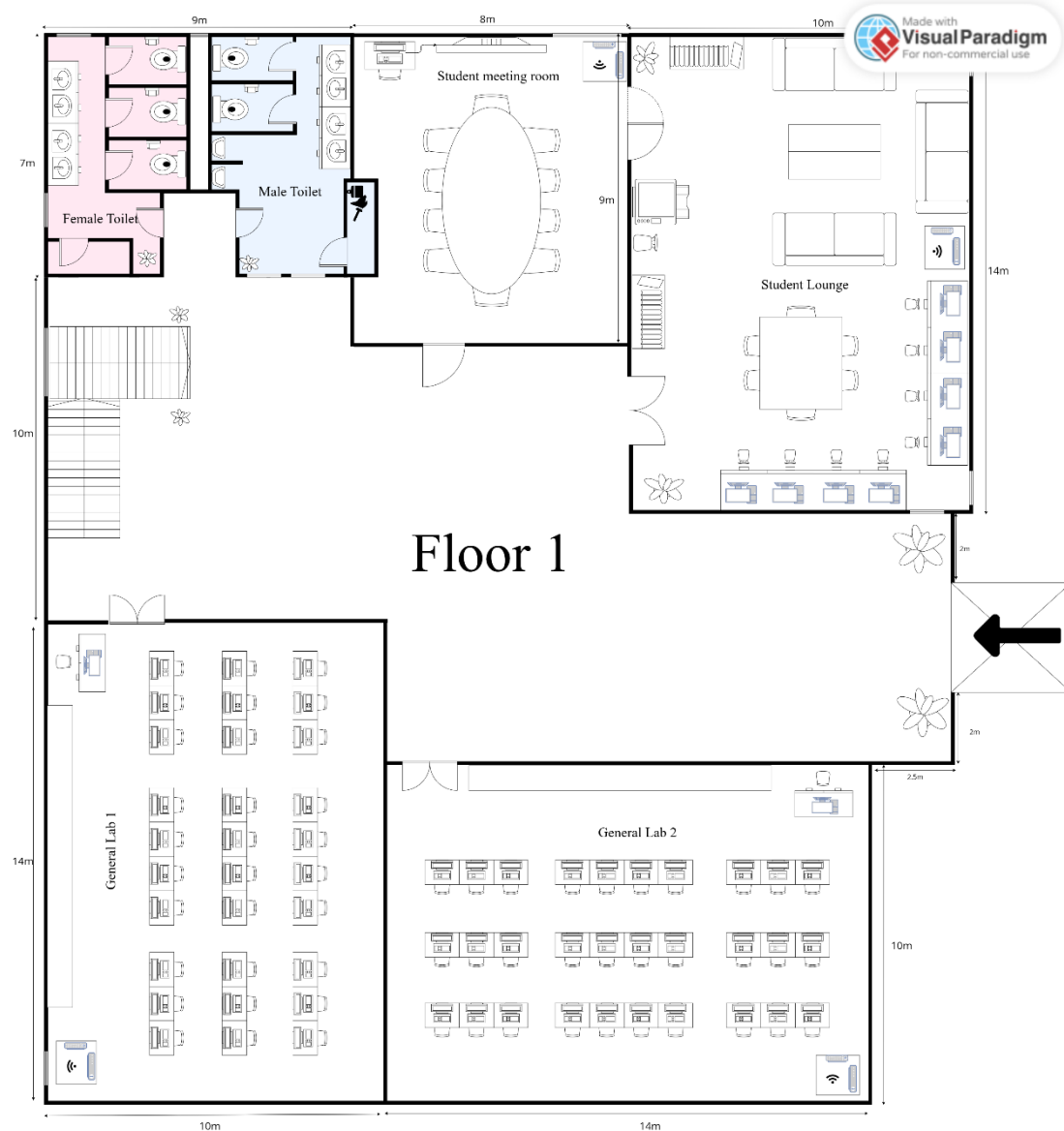


Figure 1.0

Figure 1.0 above shows the full layout of the first floor of the building. It is equipped with 2 general labs, 1 student lounge, 1 student meeting room and 2 toilets. The general labs are placed at first floor so that it's much easier for students to access. Each lab can accommodate a maximum number of 30 students and each table are equipped with a monitor and a CPU for enhanced learning experience. Both labs are also equipped with a high-bandwidth WIFI router to give a seamless and smooth network experience during lecture sessions. Both are width 10m and a length of 14m.

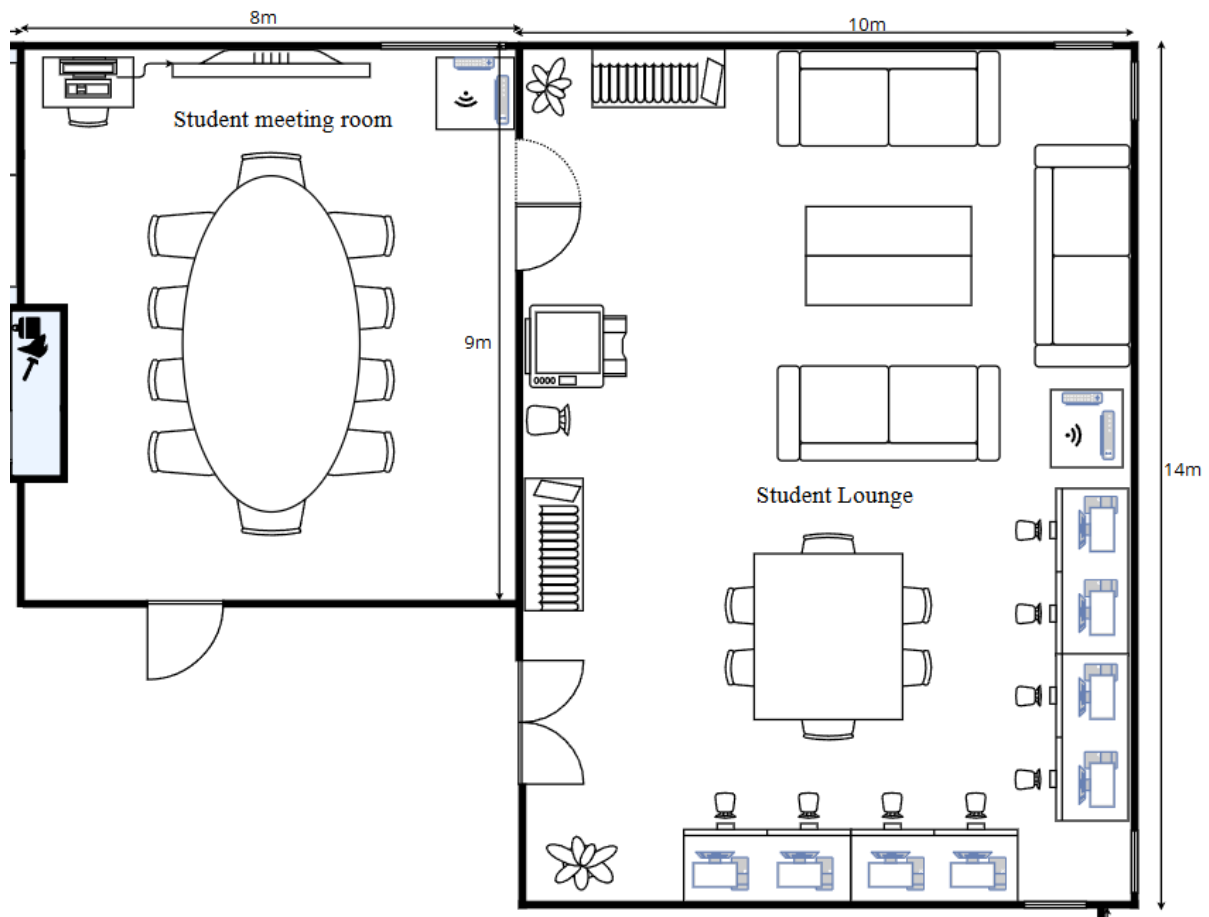


Figure 1.1

Figure 1.1 shows more details of the student lounge and meeting room. The student lounge provides students an area to work and relax while connected to the network via WIFI. The room is equipped with books and computers for students use. Beside it is a student meeting room which is accessible through the door from the hallway or student lounge. It aims to provide a suitable environment to discuss their group project or assignments. The student lounge measures 14 m by 10 m, while the meeting room is 9 m by 8m. We hope that these 2 facilities will provide a much more favourable environment for student to study or even rest.

Second Floor

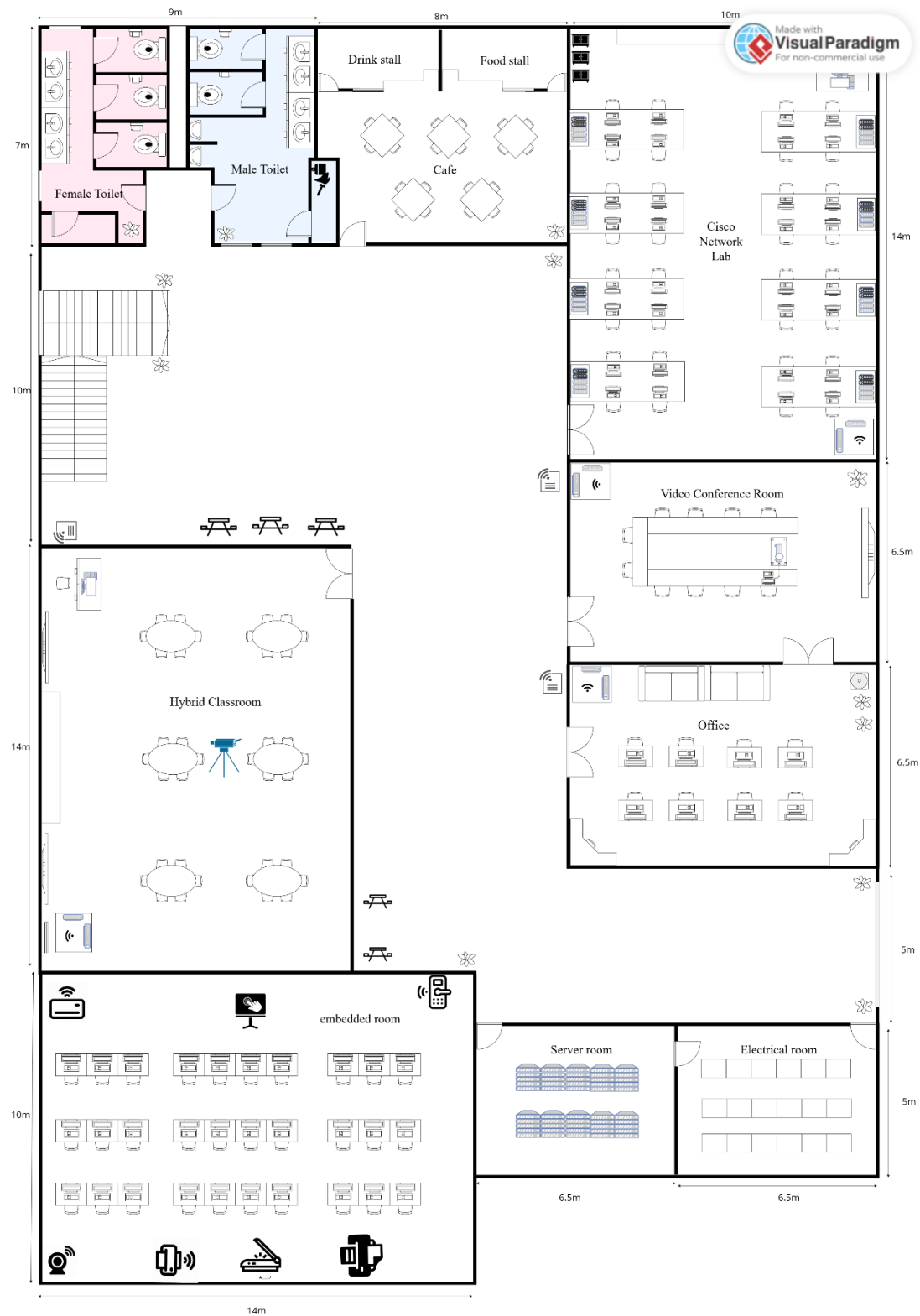


Figure 1.3

The second floor layout includes various rooms and facilities designed for accessibility and functionality. On the right side, there is a Cisco Network Lab (10m x 14m) with multiple workstations, monitors, and CPUs for student use, along with high-bandwidth Wi-Fi for seamless connectivity. Adjacent to it is the Video Conference Room (6.5m x 10m) with a large table and monitors for virtual meetings, and next to that, an Office space (6.5m x 10m) with desks and computers for staff. The bottom left features the Embedded Room (14m x 10m) with specialized equipment for embedded systems, while the bottom right houses a Server Room (6.5m x 5m) for networking equipment and an Electrical Room (6.5m x 5m) for the building's electrical infrastructure. The left side includes a Hybrid Classroom (10m x 14m) with round tables for group work and hybrid learning equipment. Near the center, a cafe area with food and drink stalls offers seating, and there's an open student lounge with additional seating. Male and female toilets are located near the cafe area for convenience.

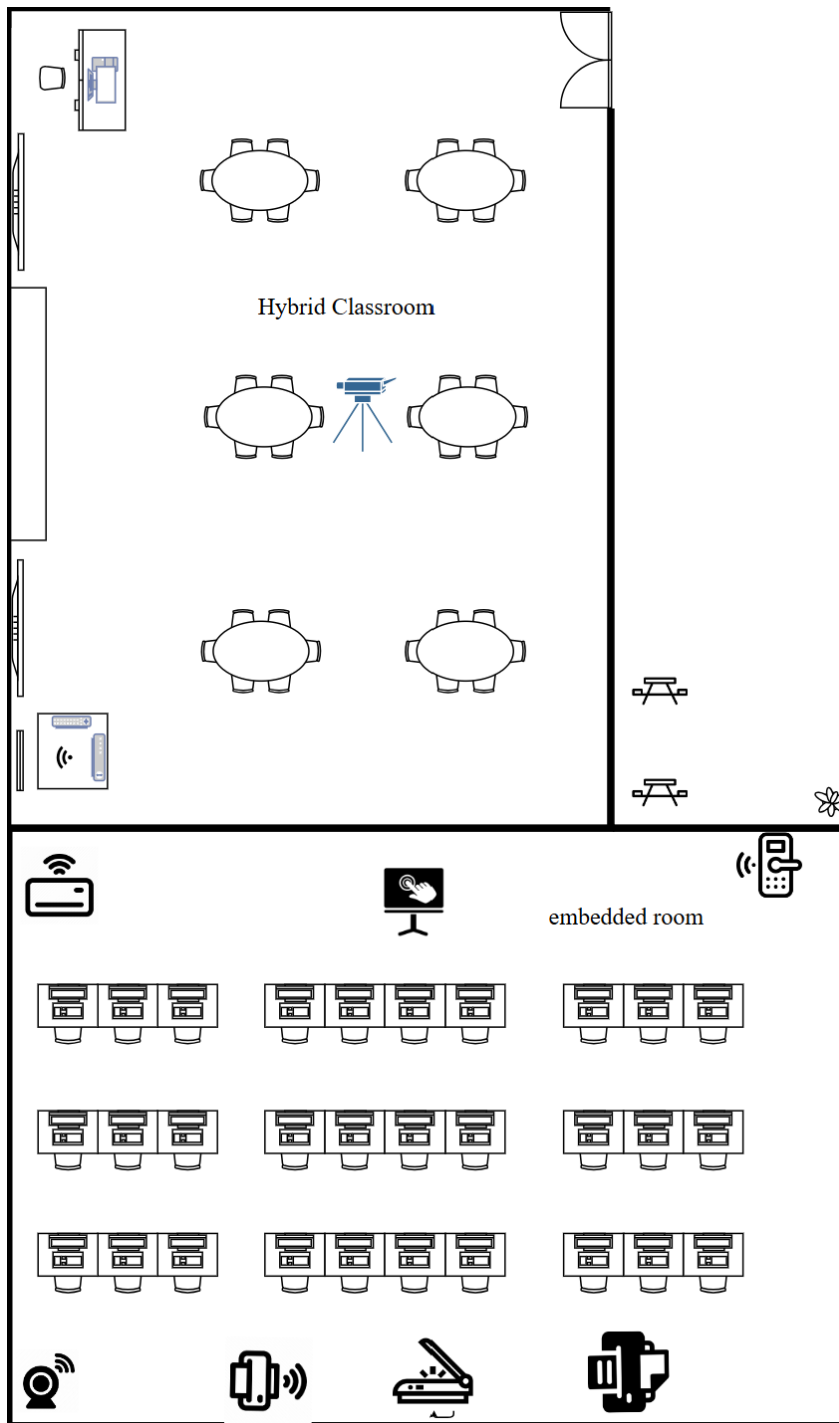


Figure 1.4

Figure 1.4 provides a detailed view of the Embedded Room and Hybrid Classroom. The Hybrid Classroom offers a conducive learning environment, featuring a 360-degree camera for monitoring students, two large screens at the front for video conferencing, and tables and chairs arranged in groups to facilitate discussions. High-speed Wi-Fi ensures smooth connectivity. The Embedded Room is equipped with IoT-enabled devices, including a wireless air conditioner, scanner, printer, and a digital camera. Access to the room is secured with a digital lock, with all technology in the room operated through IoT integration.

Brief Explanation for The Layout

This layout represents a modern academic facility tailored to the Faculty of Computing's needs, with a clear organization of spaces for both teaching and collaborative learning.

For the Central Features, the main entrance opens into a café area with drink and food stalls, providing a casual space for students and staff to relax or study informally. Nearby are the male and female restrooms, conveniently located for ease of access from different parts of the building. This café area serves as a social hub for students, fostering interaction and a sense of community.

The Academic and Collaboration space in on the right side, the layout includes a Cisco Network Lab, a dedicated space for networking activities and teaching. This lab is arranged to accommodate multiple workstations with adequate spacing, which is essential for hands-on networking exercises. Just below, there is a Video Conference Room equipped with a large table and conference technology, designed for virtual meetings and collaborations with remote participants.

The Hybrid Classroom on the left side of the layout is designed to support both in-person and remote learning. It features multiple round tables, likely intended for group discussions, collaborative projects, and interactive teaching methods. A hybrid setup, typically with a central display and conferencing equipment, is ideal for blended learning, enabling effective engagement for both on-site and remote students.

As for Administrative and Technical space which is towards the bottom of the layout, there is an Office space designated for staff or administrative work, equipped with desks and essential office furniture. Adjacent to the office, the Server Room and Electricity Room are located, centralizing the building's IT infrastructure and ensuring quick access for maintenance or troubleshooting. This setup ensures that network operations and electrical systems are secure and easily manageable, meeting the building's technical demands.

Overall, this layout efficiently combines academic, collaborative, and technical spaces to support a seamless educational experience, encouraging both learning and professional development while maintaining robust infrastructure for future growth.

Meeting Minutes

Meeting Minutes #1

Date/Time		1/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Overviewing task 1 2. Designing the floor plan 3. Deciding tools for floor plan designing 4. Task distribution for each group member	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		20 00	-
Neo Li Xin		20 00	-
Pravinraj a/l Sivabathi		20 00	
Ahmad Ziyaad bin Mohd Abbas		20 00	
Meeting details			
No.	Discussing item	Item details	Person-In charge
1.	Overviewing the case study of task 1 and make a simple discussion	- Choh Jing Yi show the case study and rubric of the task 1 - Make a simple discussion of the project and make sure everyone group member understands the project requirements and goals	Choh Jing Yi
2.	How to design the floor plan	-All group members find some floor plan template on online as inspiration for design -We also discuss about the design of the network infrastructure and project setup	All group members
3	Discuss what tools for drawing the floor plan	-We use ChatGPT to suggest some tool to designing the floor plan (AutoCAD, SketchUp, FloorPlanner and Visual Paradigm) -We try to use the tools suggested but found it is	All group members

		complicated for drawing and some required subscription payment -Lastly, we select Visual Paradigm as our drawing tool for floor planning -Then we discuss about the shape and place of each lab	
4	Task distribution for each group member	Task: Choh Jing Yi-initial floor plan layout, designing for general lab and student lounge. Oversee progress tracking and timeline management, documenting the meeting minutes and ensure all design logged Neo Li Xin- designing video conference, cisco network lab and stall. Documenting meeting minutes and review the final design layout. Pravinraj a/l Sivabathi-designing embedded lab Ahmad Ziyaad bin Mohd Abbas-designing hybrid classroom	All group members
5	Discussing the place and size of each lab and student lounge	Choh Jing Yi-suggest put general lab at the ground floor for student easier attending classes.	
6	Discussing additional rooms or equipment that can be added to the floor plan.	Choh Jing Yi-suggested adding a toilet for each floor Neo Li Xin-suggest adding some food and drink stall for student and staff Pravinraj a/l Sivabathi-recommended adding a lecturer office for lecturers to rest and complete tasks.	
7	Meeting ended and everyone start to do task distributed	-We ended the meeting at 21.30 after all the discussion had been done	All group members

Meeting Minutes #2

Date/Time		3/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Check the progress of each member 2. Edit the design and size of floor planning if found not satisfied 3. Combination of each group member drawing 4. Adding some useful appliances	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		20 00	-
Neo Li Xin		20 00	-
Pravinraj a/l Sivabathi		20 00	-
Ahmad Ziyaad bin Mohd Abbas		20 00	-
Meeting details			
No.	Discussing item	Item details	Person-In charge
1.	Check the progress of each group member	- Share the floor planning drawing on visual paradigm	Choh Jing Yi
2.	Edit the design and size of each floor	-Make sure everyone put the scale of distance for each lab and others room -Put some decoration at the corner of the room or lab	All group members
3.	Further improvisation on the floor planning and room equipment	-Choh Jing Yi suggest adding some decoration like flowers and benches -Pravinraj suggest adding electrical and server room -Neo Li Xin suggest the addition of tracking camera that follows the speaker and	Choh Jing Yi, Pravinraj, Neo Li Xin

		project it onto the smart TV in the hybrid classroom.	
4.	Final design review	- Ensure that all layouts are accurately drawn, well-connected, and neatly organized. -Ensure scale are properly documented - Ensure that room positioning and layout are logical	All group members
5.	End of meeting	-We ended the meeting at 20.40 after the building design layout is confirmed.	All group members